A. *Proof.* Let β be regular. We have that $\alpha = x \circ \beta$, and so $\dot{\alpha} = \nabla x \cdot \dot{\beta} = x_u\dot{\beta}_1 + x_v\dot{\beta}_2$. We know that $\dot{\beta}_1$ and $\dot{\beta}_2$ never equal 0 at the same time. We also know that neither x_u or x_v are 0 because x is a local parameterization. Therefore, $\dot{\alpha}$ is regular.

Next, let α be regular. Similarly, $\dot{\alpha} = \nabla x \cdot \dot{\beta}_1 = x_u\dot{\beta}_2 + x_v\dot{\beta}$. x_u and x_v are linearly independent, and the LHS can never be 0, therefore $\dot{\beta}_1$ and $\dot{\beta}_2$ are never 0 at the same time. Therefore, β is regular □

B. Below is my code to find the matrix g.

```
In[94]:= Clear[x, xu, xv, g]
         x = {2 u, 2 v, u^2 + v^2 - 1} / (u^2 + v^2 + 1);
         xu = D[x, u];
         xv = D[x, v];
         g = {{Dot[xu, xu], Dot[xu, xv]}, {Dot[xu, xv], Dot[xv, xv]}};
         Simplify[g]
Out[99]= {{4/(1 + u^2 + v^2)^2, 0}, {0, 4/(1 + u^2 + v^2)^2}}
```

Figure 1: My code.

So, $g = \frac{4}{(1+u^2+v^2)^2}\mathbb{1}$. This is a scalar of the identity matrix.

C. (a) We compute that $x_u = (\cos v \cos u, \sin v \cos u, -\sin u)$. We also compute that $x_v = (-\sin u \sin v, \sin u \cos v, 0)$. Then, $x_u \cdot x_u = \cos^2 u \cos^2 v + \cos^2 u \sin^2 v + \sin^2 u = 1$. Also, $x_u \cdot x_v = \cos v \cos u \sin u \sin v - \cos v \cos u \sin u \sin v = 0$. Finally, $x_v \cdot x_v = \sin^2 u \sin^2 v + \sin^2 u \cos^2 v = \sin^2 u$. So,

$$g = \begin{bmatrix} 1 & 0 \\ 0 & \sin^2 u \end{bmatrix}$$

.

(b) I could have used the fact that $J^T J = g$, where J is the Jacobian.

(c) Let x_1 be the quasi-inverse of the stereographic project of $\mathbb{S}^2$ and x_2 be the parameterization of $\mathbb{S}^2$ using spherical coordinates. x_1 sends the point $(1, 0)$ to $(1, 0, 0)$, and at that point $g_1 = \frac{4}{(1+1)^2}\mathbb{1} = \mathbb{1}$. x_2 sends the point $(\frac{\pi}{2}, 0)$ to $(1, 0, 0)$, and $g_2 = \begin{bmatrix} 1 & 0 \\ 0 & \sin^2(\frac{\pi}{2}) \end{bmatrix} = \mathbb{1}$. So, they do have the same g matrix at $(1, 0, 0)$.

(d) x_1 sends the point $(\sqrt{2}+1, 0)$ to $(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})$. At that point $g_1 = \frac{4}{(2+\sqrt{2})^2}\mathbb{1}$. x_2 sends the point $(\frac{\pi}{4}, 0)$ to $(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})$. At that point $g_2 = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$. By observation, $g_1 \neq g_2$. So, x_1 and x_2 have different g matrices at $(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})$.